

A Brief Overview of Web API

What is Web API?

A web API (Application Programming Interface) is a set of rules and specifications that allows different software applications to communicate and exchange data, particularly over the internet. It acts as an intermediary, enabling a client (like a web browser, mobile app, or another server) to request and receive data from a server application.

Layers in Web API:

1. Presentation Layer:

The presentation layer, also known as the API endpoint or the public-facing interface, is responsible for handling incoming requests and returning responses. It's the layer that developers interact with most directly, providing a way to access the functionality and data of the underlying service.

Responsibilities:

- *.Handling HTTP Requests
- *.Data Validation and Formatting
- *.Returning Responses
- *.Error Handling

2. API Layer:

It serves as the entry point for client applications to interact with the server-side logic and data. It acts as a bridge, handling incoming requests, validating input, and formatting responses in a way that clients can understand.

Responsibilities:

- *.Exposing Endpoints
- *.Handling Requests
- *.Input Validation
- *.Data Transformation

3. Database Layer:

It is responsible for managing the persistent storage and retrieval of data. It's a crucial component that enables the API to interact with databases, ensuring data is stored, updated, and accessed efficiently.

Responsibilities:

- *.Data Storage
- *.Data Retrieval
- *.Data Manipulation
- *.Integration with Other Layers

HTTP Verbs:

HTTP verbs, often referred to as HTTP methods, are a fundamental part of the HTTP protocol used in Web APIs to indicate the desired action to be performed on a resource. They form the backbone of RESTful API design by helping define standard operations on web resources.

1.GET - Pull the data from database.

It is used to retrieve data from a server without making any changes to the resource.

Features:

- *.Safe
- *.Idempotent
- *.Read-Only Operation
- *.Cacheable

2.POST - Insert data into database.

Creates a new resource on the server. It's used for creating new data, such as submitting a form or creating a new item in a database.

Features:

- *.Used to Create Resources
- *.Changes Server State
- *.Returns Created Resource or Status

3.PUT - Update data in database.

Replaces an entire resource with new data. The server replaces the existing resource with the provided data, so the updated version of the resource is entirely replaced.

Features:

- *.Update or Create Resource
- *.URI-Specific
- *.May Return Different Status Codes

4.DELETE - Delete data from database.

It is used to request that the server delete a specific resource. It tells the server to remove the resource identified by the URL in the request, effectively "getting rid of it" from the server's perspective. This is often used to remove data, like deleting a user account or a product from a database.

Features:

- *.No Request Body
- *.Returns (Typical Response)
- *.Idempotent